

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410677
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Roselier; Samuel et al.

Multiple expandable metal packers with hydrolock prevention

Abstract

A technique facilitates improved actuation and use of packers disposed along a well string and placed in a borehole, e.g. a wellbore. A plurality of packers may be disposed along a base pipe, e.g. a production tubing. Each packer has an expandable sleeve which may be expanded into a fully expanded configuration establishing a secure, sealing engagement with a surrounding wall. e.g. a surrounding wellbore wall. Additionally, a hydrolock prevention valve system comprises a valve which may be actuated to allow or block the flow of fluid from an annular region or regions between the packers.

Inventors:	Roselier; Samuel (Bruz, FR), Neveu; Romain (Bruz, FR), Laupie; Robin (Bruz, FR), Saltel; Nicolas (Rio de Janeiro, BR), Patel; Dinesh (Humble, TX)
Applicant:	Schlumberger Technology Corporation (Sugar Land, TX)
Family ID:	87349248
Assignee:	Schlumberger Technology Corporation (Sugar Land, TX)
Appl. No.:	18/720970
Filed (or PCT Filed):	January 23, 2023
PCT No.:	PCT/US2023/011307
PCT Pub. No.:	WO2023/141311
PCT Pub. Date:	July 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250075588 A1	Mar. 06, 2025

Related U.S. Application Data

us-provisional-application US 63302222 20220124

Publication Classification

Int. Cl.: E21B33/122 (20060101); E21B34/06 (20060101)

U.S. Cl.:

CPC E21B33/122 (20130101); E21B34/06 (20130101);

Field of Classification Search

CPC: E21B (33/1243); E21B (33/127); E21B (33/1272); E21B (33/1275)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3456725	12/1968	Hatch	166/147	E21B 33/14
3948145	12/1975	Garaty	91/519	E21B 19/086
5549165	12/1995	Brooks	166/185	E21B 33/1243
6047239	12/1999	Berger	702/9	E21B 49/08
9217308	12/2014	Wood	N/A	N/A
10190386	12/2018	Stæhr	N/A	N/A
10947810	12/2020	Vasques	N/A	N/A
11091975	12/2020	Roselier	N/A	N/A
11142987	12/2020	Vasques	N/A	N/A
2002/0100585	12/2001	Spiers	166/264	E21B 49/0813
2006/0248949	12/2005	Gregory	N/A	N/A
2009/0183882	12/2008	van Zuilekom	166/371	E21B 49/081
2014/0190708	12/2013	Hallundbæk	N/A	N/A
2021/0071495	12/2020	Vasques	N/A	E21B 33/1208

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3159478	12/2016	EP	N/A
3628813	12/2019	EP	N/A
3601718	12/2020	EP	N/A
2020152262	12/2019	WO	N/A

OTHER PUBLICATIONS

Search Report and Written Opinion of International Patent Application No. PCT/US2023/011307 dated May 11, 2023, 8 pages. cited by applicant

Primary Examiner: Carroll; David

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage entry under 35 U.S.C. 371 of International Application No. PCT/US2023/011307, entitled “Multiple Expandable Metal Packers with Hydrolock Prevention,” filed Jan. 23, 2023, which claims the benefit of U.S. Provisional Application No. 63/302,222, entitled “Multiple Expandable Metal Packers with Hydrolock Prevention,” filed Jan. 24, 2022, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) In many oil and gas well applications, a wellbore is drilled into the earth and through a reservoir of a desired fluid, e.g. oil and/or gas. The wellbore may subsequently be completed with appropriate completion equipment having packers which may be expanded to isolate regions along the wellbore. For example, packers may be disposed along sand control equipment or other types of completion equipment to facilitate production of the desired fluids from the reservoir. Depending on the application, the packers may be mounted along a well string and selectively expanded to effectively form a seal between base pipe of the well string and the surrounding wellbore wall. In some applications, multiple packers are expanded simultaneously and fluid becomes trapped between the packers. As expansion of the packers continues, the trapped fluid can prevent the packers from expanding properly to their fully expanded configuration.

SUMMARY

(2) In general, a system and methodology facilitate improved actuation and use of packers disposed along a well string and placed in a borehole, e.g. a wellbore. A plurality of packers may be disposed along a base pipe, e.g. a production tubing. Each packer has an expandable sleeve which may be expanded into a fully expanded configuration which establishes a secure, sealing engagement with a surrounding wall, e.g. a surrounding wellbore wall. Additionally, a hydrolock prevention valve system comprises a valve which may be actuated to allow or block the flow of fluid from an annular region or regions between the packers.

(3) However, many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein, and:

(2) FIG. 1 is an illustration of an example of a well system having multiple packers positioned along a well string deployed in a borehole, e.g. a wellbore, in combination with a hydrolock prevention valve system according to an embodiment of the disclosure;

(3) FIG. 2 is a cross-sectional illustration of an example of a hydrolock prevention valve system, according to an embodiment of the disclosure;

(4) FIG. 3 is a cross-sectional illustration of the hydrolock prevention valve system shown in FIG. 2 but in a different operational position, according to an embodiment of the disclosure;

(5) FIG. 4 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system deployed in a run-in-hole configuration, according to an embodiment of the disclosure;

- (6) FIG. 5 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system deployed downhole in another configuration, according to an embodiment of the disclosure;
- (7) FIG. 6 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system deployed downhole in another configuration, according to an embodiment of the disclosure;
- (8) FIG. 7 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system deployed downhole in another configuration, according to an embodiment of the disclosure;
- (9) FIG. 8 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system combined with an expansion valve system, according to an embodiment of the disclosure;
- (10) FIG. 9 is an illustration of an example of a well system having multiple packers and a hydrolock prevention valve system combined with both an expansion valve system and a compensation valve system, according to an embodiment of the disclosure;
- (11) FIG. 10 is another illustration of an example of a well system having multiple packers and a hydrolock prevention valve system combined with both an expansion valve system and a compensation valve system, according to an embodiment of the disclosure; and
- (12) FIG. 11 is an illustration of another example of a well system having multiple packers and a hydrolock prevention valve system which may be utilized with individually expandable packers, according to an embodiment of the disclosure.
- (13) FIG. 12 is an illustration of another example of a well system having multiple packers and a hydrolock prevention valve system which may be utilized with individually expandable packers, according to an embodiment of the disclosure.
- (14) FIG. 13 is an illustration of another example of a well system having multiple packers and a hydrolock prevention valve system which may be utilized with individually expandable packers, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

- (15) In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that the system and/or methodology may be practiced without these details and that numerous variations or modifications from the described embodiments may be possible.
- (16) The disclosure herein generally involves a system and methodology which facilitate improved actuation and use of multiple packers disposed in a wellbore or other type of borehole. According to an embodiment, a well system comprises a plurality of packers disposed along a base pipe, e.g. a production tubing, of an overall well string. In some embodiments, the plurality of packers may comprise two packers which may be arranged as straddle packers or other types of cooperating pairs of packers. However, other embodiments may comprise three or more packers arranged along the well string.
- (17) According to an embodiment of the well system, each packer may have an expandable sleeve which may be expanded into a fully expanded configuration which establishes a secure, sealing engagement with a surrounding wall, e.g. a surrounding wellbore wall. By way of example, the expandable sleeve may be an expandable metal sleeve of an expandable metal packer. In an expandable metal packer, the expandable sleeve may be in the form of a metal bladder which is selectively deformed downhole by, for example, hydroforming.
- (18) Additionally, the well system may comprise a hydrolock prevention valve system having a valve which may be actuated to allow or block the flow of fluid from an annular region or regions between the packers. For example, the valve may be set to allow flow from the annular region(s) between the packers to an annular region outside of the packers, e.g. an annular region uphole or downhole of the packers. The valve may then be selectively actuated to a closed position when, for

example, the packers are fully set in their fully expanded configuration. In some embodiments, the actuation of the valve to the closed position may be automatically triggered when a predetermined pressure is reached within the expanded packers.

(19) Referring generally to FIG. 1, an example of a well string **20** is illustrated as positioned within a borehole **22**, e.g. a wellbore, and comprising a well system **24**. In this embodiment, the well string **20** comprises multiple packers **26** disposed along a base pipe **28**. FIG. 1 illustrates the multiple packers **26** as comprising two packers **26** although three or more packers **26** may be used in other embodiments.

(20) Each packer **26** may be in the form of an expandable packer with an expandable sleeve **30**. By way of example, each packer **26** may be an expandable metal packer in which the expandable sleeve **30** is in the form of an expandable metal sleeve which may be constructed as a metal bladder **32**. The metal sleeve/bladder **32** may be expanded and deformed downhole under fluid pressure, e.g. by hydroforming. In some embodiments, each expandable sleeve **30** is secured to base pipe **28** by pipe connectors **34** which may be in the form of packer extremities connected to the longitudinal extremity of each expandable sleeve **30**. When fluid under sufficient pressure is directed into a packer cavity **35** between the pipe connectors **34** of each packer **26** and between the corresponding expandable sleeve **30** and base pipe **28**, the expandable sleeve **30** is deformed in a radially outward direction. If sufficient pressure is continually applied, the expandable sleeve **30** deforms into secure, sealing engagement with a surrounding wall **36**, e.g. a surrounding wellbore wall, as illustrated in FIG. 1.

(21) As illustrated, the well string **20** also comprises a hydrolock prevention valve system **38** which has a valve **40**, a first flow passage **42** extending to an annular space **44** located between packers **26**, and a second flow passage **46** extending to an annular section **48** outside of the packers **26**. In other words, the second flow passage **46** may extend to annular section **48** located uphole or downhole of the plurality of packers **26**. The valve **40** is selectively actuatable to allow or block flow between the first flow passage **42** and the second flow passage **46**.

(22) Referring generally to FIGS. 2 and 3, one embodiment of hydrolock prevention valve system **38** is illustrated. In this embodiment, valve **40** is slidably positioned in a valve housing **50** between first flow passage **42** and second flow passage **46**. The valve **40** may comprise an elongated piston **52** slidably mounted in a corresponding piston passage **54** of valve housing **50**. Furthermore, the elongated piston **52** may have seals **56** which form a sealing engagement between elongated piston **52** and the surrounding wall forming piston passage **54**. The seals **56** are arranged to enable selective flow of fluid between first flow passage **42** and second flow passage **46** via connection through passage **54** or to block flow of fluid therethrough (depending on the position of piston **52**).

(23) In this example, the valve housing **50** also comprises a tubing passage **58** in fluid communication with piston passage **54** and with the interior of base pipe **28** via one or more expansion ports **60** extending laterally through the wall of base pipe **28** (see FIG. 1). The valve housing **50** also may comprise a packer passage **62** in fluid communication with piston passage **54** and with the packer cavities **35**. The packer passage **62** may be placed in fluid communication with packer cavities **35** via ports/passages **64** formed through at least some of the pipe connectors/extremities **34** (see FIG. 1).

(24) In FIG. 2, the valve **40** is illustrated in the open position allowing flow of fluid between the first flow passage **42** and the second flow passage **46**. The valve **40** may be placed in this configuration when, for example, the well string **20** is run-in-hole and during setting of packers **26**. In this valve configuration, the interior of the base pipe **28** is in fluid communication with the packer cavities **35** via tubing passage **58** and packer passage **62** to allow expansion of the expandable sleeves **30** when fluid under sufficient pressure is directed down through the interior of base pipe **28**. In this valve position, the annular space **44** between packers **26** also is connected to the annulus externally of packers **26**, i.e. to annular section **48**.

(25) As a result, liquid in the annular space **44** is able to bleed out through first flow passage **42**,

piston passage **54**, and second flow passage **46** to the annular section **48**. Because the liquid is able to bleed from annular space **44**, the potential for hydrolock inside this annular space **44** during expansion of packers **26** is reduced or eliminated. In other words, hydrolock is not able to prevent the packers **26** from fully expanding.

(26) Once the packers **26** are fully set via expansion to their fully expanded configuration, the valve **40** may be actuated to the closed position blocking flow between first flow passage **42** and second flow passage **46**, as illustrated in FIG. **3**. In this flow blocking position, piston **52** has been shifted so that seals **56** prevent fluid flow between first and second flow passages **42**, **46**. In some embodiments, the piston **52** is initially held in the position allowing flow by a securing element **65**, such as a shear element **66** which ruptures at a preset pressure level. The shear element **66** may be constructed so the preset pressure level is high enough to allow full expansion of the packers **26** while the piston **52** is exposed to the expansion pressure via passages **58**, **62**. However, once the pressure rises above the preset pressure level, the shear element **66** is sheared and the piston **52** is shifted to the flow blocking position illustrated in FIG. **3**. In some embodiments, the piston **52** may be held in this flow blocking position by a mechanical device **68**, such as a spring-loaded retention member **70**.

(27) In operation, the well string **20** and well system **24** are initially deployed down into borehole **22** with packers **26** in a contracted or unexpanded configuration, as illustrated in FIG. **4**. Once the packers **26** are at a desired location in borehole **22**, fluid under pressure may be directed down through the interior of base pipe **28**, through lateral port **60**, through pipe connector ports **64**, and into cavities **35** to initiate expansion of expandable packer sleeves **30**, as illustrated in FIG. **5** by arrows **72**.

(28) Initially, the sleeves **30** are expanded into first contact with the surrounding borehole wall **36**. In some embodiments, the sleeves **30** may be metal sleeves comprising metal bladders **32** with a sealing element along the outer surface to facilitate creation of a seal with the surrounding wall **36**. Once the seal is initiated, the confined annular space **44** is established. As expansion of the packers **26** is continued to the fully expanded configuration illustrated in FIG. **6**, the annular space **44** is further constrained.

(29) However, as annular space **44** decreases in size excess fluid within the confined annular space **44** is able to escape via hydrolock prevention valve system **38** by flowing through first flow passage **44**, valve **40**, and second flow passage **46** to annular section **48**, as represented by arrows **74**. It should be noted the first flow passage **44** may be established via passages extending through the appropriate pipe connectors **34** and packer cavity(ies) **35** as illustrated. After the packers **26** have been deformed/expanded to their fully expanded configuration and excess fluid/pressure has been bled from annular space **44**, the valve **40** may be transitioned to the flow blocking position as illustrated in FIG. **7**. Actuating valve **40** to the closed or flow blocking position may be achieved via increasing the pressure within packers **26** above a predetermined level to cause shifting of piston **52** as described above.

(30) Referring generally to FIG. **8**, another embodiment of well system **24** is illustrated. In this example, an expansion valve system **76** is used in cooperation with the hydrolock prevention valve system **38**. The expansion valve system **76** comprises an expansion valve **78** which allows communication between the interior of base pipe **28** and the expandable packers **26** during expansion of the packers **26**. However, the expansion valve **78** closes off the lateral expansion port **60** at a preset pressure. The expansion valve system **76** and the hydrolock prevention valve system **38** may be combined in the same valve housing or they may use separate valve housings. Depending on the application, the two valve systems **38**, **76** may utilize a common piston **52** or separate pistons.

(31) As illustrated in FIG. **9**, the hydrolock prevention valve system **38** also may be combined with a compensation valve system **80** or with both the expansion valve system **76** and the compensation valve system **80**. The compensation valve system **80** is used for annulus pressure compensation

when the packers **26** are set. For example, the well system **24** may utilize two compensation valve systems **80** with one system **80** placed on each end of the plurality of packers **26**. This enables annulus pressure compensation with respect to an uphole annulus section (as represented by arrows **82** in FIG. **9**) and/or with respect to a downhole annulus section (as represented by arrows **84** in FIG. **10**). In this latter embodiment, three packers **26** are illustrated although other numbers of packers **26**, e.g. greater numbers of packers **26**, also may be utilized in a given well system **24**. (32) In embodiments described above, the packers **26** are expanded to their fully expanded/set configuration simultaneously. However some embodiments may be constructed so the packers **26** can be set independently, as illustrated in FIG. **11**. As illustrated, two of the packers **26** have been expanded while one of the packers **26** has not yet been expanded. The independent and individual expansion of packers **26** may be achieved via a variety of tools, such as a straddle cup tool which allows pressure to be applied in a specified zone for inflation and expansion of specified corresponding packers **26**.

(33) Depending on the parameters of a given operation and the environment in which such operation is conducted, the well system **24** may comprise a variety of components and configurations. For example, the packers **26** and expandable sleeves **30** may be made in a variety of sizes, configurations, and materials. Similarly, the valve systems **38**, **76**, **84** may be constructed in various cooperating combinations and with various pistons, flow passage architectures, seals, and/or other components and features to accommodate the desired actuations. Similarly, the packers **26** and overall construction of well system **24** may be selected for use in open hole boreholes, cased boreholes, or within other types of downhole tubing.

(34) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A system for use in a well, comprising: a well string size for deployment in a wellbore, the well string comprising: a base pipe; a plurality of packers disposed along the base pipe, each packer having an expandable metal sleeve which may be expanded into a fully expanded configuration forming a secure, sealing engagement with a surrounding wall; an annular space between packers of the plurality of packers; and a hydrolock prevention valve system having a valve, a flow passage within a valve body between the annular space and an annular section outside of the plurality of packers, the valve being actuatable such that a piston of the valve allows flow between the annular space and the annular section outside of the plurality of packers in a first position and blocks flow between the annular space and the annular section outside of the plurality of the packers in a second position, wherein the piston includes a tapered outer surface and a groove, wherein the valve body includes a spring-loaded retention member. and wherein in response to the piston moving to the second position, the spring-loaded retention member moves along the tapered outer surface and into the groove, thereby retaining the piston in the second position.
2. The system as recited in claim 1, wherein the valve is configured to allow flow between the annular space and the annular section outside of the plurality of the packers while the well string is run into the wellbore.
3. The system as recited in claim 1, wherein the valve is configured to allow flow between the annular space and the annular section outside of the plurality of the packers during expansion of the plurality of packers into the fully expanded configuration so as to enable evacuation of excess fluid from the annular space between packers.
4. The system as recited in claim 1, wherein the valve is configured to block flow between the annular space and the annular section outside of the plurality of the packers after expansion of the

plurality of packers to the fully expanded configuration.

5. The system as recited in claim 1, wherein the plurality of packers comprises two packers.

6. The system as recited in claim 1, wherein the plurality of packers comprises at least three packers.

7. The system as recited in claim 1, further comprising an expansion valve configured to work in cooperation with the hydrolock prevention valve system.

8. The system as recited in claim 1, further comprising a compensation valve configured to work in cooperation with the hydrolock prevention valve system.

9. The system as recited in claim 7, further comprising a compensation valve configured to work in cooperation with the hydrolock prevention valve system.

10. The system as recited in claim 1, wherein the packers of the plurality of packers are configured to be expanded simultaneously.

11. The system as recited in claim 1, wherein selected packers of the plurality of packers are configured to be expanded individually.
